

ChE 311H - Fall 2023 HW #5

Due date: Friday, October 27, 2023. Bring to Dr. Khan's office before 5pm. You can slip it under his door. Submit as a hardcopy.

Instructions: Show all work and box in your final answer. Write all group member's names on the first page and the group number in top corner of all subsequent pages. Remember, illegible solutions will receive no credit. Additionally, regrade requests must be submitted within 1 week of receiving your grade.

1. BSL 6A.2

2. BSL 6A.4

3. BSL 7A.2

4. BSL 7A.5

5. A closed tank contains compressed air and oil ($SG_{oil} = 0.90$) as is shown in Fig.2. A U-tube monometer using mercury ($SG_{Hg} = 13.6$) is connected to the tank as shown. The column heights are $h_1 = 36$ in., $h_2 = 6$ in., and $h_3 = 9$ in. Find the pressure reading (in psi or kPa) of the gage (this is the same as the pressure of the air in tank).

